In [ ]:

```
EXPERIMENT 8
DATA PRE-PROCESSING & FEATURE ENGINEERING

PIN Number:`241UDS0103`

### AIM
To implement data pre-processing and feature-engineering techniques on a
dataset, including removing unnecessary features, scaling numerical
features, and encoding categorical variables using Python libraries.
```

In [ ]:

```
## Step 1: Import the Basic Libraries
```

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:

```
## Step 2: Load & Analyze the Dataset
```

In [2]:

```
df_241UDS0103 =
pd.read_csv("Experiment_8_241UDS0103_Student_Feature_Data.csv")
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 220 entries, 0 to 219
Data columns (total 23 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Branch               218 non-null    str
 1   Marks_10th           218 non-null    float64
 2   Marks_12th           218 non-null    float64
 3   Gender               219 non-null    str
 4   Board_10th           219 non-null    str
 5   Board_12th           219 non-null    str
 6   Category             219 non-null    str
 7   GPA_1                219 non-null    float64
 8   Normalized_Rank      218 non-null    float64
 9   CGPA                 219 non-null    float64
 10  Current_Backlogs     219 non-null    float64
 11  Ever_Backlogs        219 non-null    float64
 12  GPA_2                220 non-null    float64
 13  GPA_3                220 non-null    float64
 14  GPA_4                220 non-null    float64
 15  GPA_5                220 non-null    float64
 16  GPA_6                220 non-null    float64
 17  Olympiads_Qualified  220 non-null    int64
 18  Technical_Projects   220 non-null    int64
 19  Tech_Quiz            220 non-null    int64
 20  Engg_Coaching        220 non-null    int64
 21  Scholarships         220 non-null    int64
 22  Misc_Tech_Events     220 non-null    int64
dtypes: float64(12), int64(6), str(5)
memory usage: 39.7 KB
```

In [3]:
```
df_241UDS0103.head()
```

Out[3]: Columns 1–8 of 22

	Branch	Marks_10th	Marks_12th	Gender	Board_10th	Board_12th	Category
0	NaN	100.00	67.19	Female	CBSE	ISC	EWS
1	IT	69.70	81.03	Female	CBSE	ISC	GEN
2	ECE	90.22	71.77	Female	State Board	State Board	GEN
3	MECH	NaN	81.04	Male	State Board	State Board	EWS
4	CSE	97.77	85.35	Male	CBSE	State Board	OBC

Columns 9–15 of 22 · first column repeated

	GPA_1	Normalized_Rank	CGPA	...	GPA_3	GPA_4	GPA_5
0	9.63	4784.0	7.45	...	7.56	8.47	9.79
1	8.59	25171.0	9.43	...	6.61	9.69	6.81
2	6.16	48407.0	6.73	...	6.53	9.40	9.81
3	8.71	68285.0	6.32	...	8.72	7.26	6.70
4	6.91	19904.0	9.57	...	9.80	9.08	5.30

Columns 16–22 of 22 · first column repeated

	GPA_6	Olympiads_Qualified	Technical_Projects	Tech_Quiz	Engg_Coaching	Scholarships	Misc_Tech_Events
0	9.54	0	4	2	1	0	5
1	5.25	3	2	0	0	3	4
2	6.63	3	0	0	1	3	3
3	5.46	3	1	5	1	1	6
4	8.94	3	6	4	2	3	3

5 rows × 23 columns

In [4]:
```
df_241UDS0103.shape
```

Out[4]: `(220, 23)`

In [5]:
```
df_241UDS0103.isnull().sum()
```

Out[5]:
```
Branch                 2
Marks_10th             2
Marks_12th             2
Gender                 1
Board_10th             1
Board_12th             1
Category               1
GPA_1                  1
Normalized_Rank        2
CGPA                   1
Current_Backlogs       1
Ever_Backlogs          1
GPA_2                  0
GPA_3                  0
GPA_4                  0
GPA_5                  0
GPA_6                  0
Olympiads_Qualified    0
Technical_Projects     0
Tech_Quiz              0
Engg_Coaching          0
Scholarships           0
Misc_Tech_Events       0
dtype: int64
```

In [6]:
```
df_241UDS0103 = df_241UDS0103.dropna(how="all")
df_241UDS0103.shape
```

Out[6]:
```
(220, 23)
```

In [ ]:
```
## Step 3: Remove Unnecessary Features
```

In [7]:
```
df_241UDS0103 = df_241UDS0103.drop("Normalized_Rank", axis=1)
df_241UDS0103.shape
```

Out[7]:
```
(220, 22)
```

In [8]:
```
df_241UDS0103.columns
```

Out[8]:
```
Index(['Branch', 'Marks_10th', 'Marks_12th', 'Gender', 'Board_10th',
       'Board_12th', 'Category', 'GPA_1', 'CGPA', 'Current_Backlogs',
       'Ever_Backlogs', 'GPA_2', 'GPA_3', 'GPA_4', 'GPA_5', 'GPA_6',
       'Olympiads_Qualified', 'Technical_Projects', 'Tech_Quiz',
       'Engg_Coaching', 'Scholarships', 'Misc_Tech_Events'],
      dtype='str')
```

In [ ]:
```
## Step 4: Scale the Numerical Features
```

In [9]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
```

In [10]:
```
z_scaler_241UDS0103 = StandardScaler()
df_241UDS0103[["Marks_10th", "Marks_12th"]] =
z_scaler_241UDS0103.fit_transform(
    df_241UDS0103[["Marks_10th", "Marks_12th"]]
)
df_241UDS0103.head(3)
```

Out[10]:

Columns 1–8 of 22

	Branch	Marks_10th	Marks_12th	Gender	Board_10th	Board_12th	Category
0	NaN	2.021525	-1.217335	Female	CBSE	ISC	EWS
1	IT	-1.441601	0.275959	Female	CBSE	ISC	GEN
2	ECE	0.903724	-0.723167	Female	State Board	State Board	GEN

Columns 9–15 of 22 · first column repeated

	GPA_1	CGPA	Current_Backlogs	...	GPA_3	GPA_4	GPA_5
0	9.63	7.45	3.0	...	7.56	8.47	9.79
1	8.59	9.43	1.0	...	6.61	9.69	6.81
2	6.16	6.73	1.0	...	6.53	9.40	9.81

Columns 16–22 of 22 · first column repeated

	GPA_6	Olympiads_Qualified	Technical_Projects	Tech_Quiz	Engg_Coaching	Scholarships	Misc_Tech_Events
0	9.54	0	4	2	1	0	5
1	5.25	3	2	0	0	3	4
2	6.63	3	0	0	1	3	3

3 rows × 22 columns

In [ ]:
```
### Apply Normalization using MinMaxScaler
```

In [11]:
```
mm_scaler_241UDS0103 = MinMaxScaler()
df_241UDS0103[["CGPA", "GPA_1"]] = mm_scaler_241UDS0103.fit_transform(
    df_241UDS0103[["CGPA", "GPA_1"]]
)
df_241UDS0103.head(3)
```

Out[11]: Columns 1–8 of 22

	Branch	Marks_10th	Marks_12th	Gender	Board_10th	Board_12th	Category
0	NaN	2.021525	-1.217335	Female	CBSE	ISC	EWS
1	IT	-1.441601	0.275959	Female	CBSE	ISC	GEN
2	ECE	0.903724	-0.723167	Female	State Board	State Board	GEN

Columns 9–15 of 22 · first column repeated

	GPA_1	CGPA	Current_Backlogs	...	GPA_3	GPA_4	GPA_5
0	0.964835	0.466507	3.0	...	7.56	8.47	9.79
1	0.736264	0.940191	1.0	...	6.61	9.69	6.81
2	0.202198	0.294258	1.0	...	6.53	9.40	9.81

Columns 16–22 of 22 · first column repeated

	GPA_6	Olympiads_Qualified	Technical_Projects	Tech_Quiz	Engg_Coaching	Scholarships	Misc_Tech_Events
0	9.54	0	4	2	1	0	5
1	5.25	3	2	0	0	3	4
2	6.63	3	0	0	1	3	3

3 rows × 22 columns

In [ ]:
```
## Step 5: Encode the Categorical Features
```

In [12]:
```
from sklearn.preprocessing import LabelEncoder
```

In [13]:
```
encoder_241UDS0103 = LabelEncoder()
for col in ["Gender", "Category", "Branch"]:
    df_241UDS0103[col] =
encoder_241UDS0103.fit_transform(df_241UDS0103[col])
df_241UDS0103.head(3)
```

Out[13]:

Columns 1–8 of 22

	Branch	Marks_10th	Marks_12th	Gender	Board_10th	Board_12th	Category
0	6	2.021525	-1.217335	0	CBSE	ISC	0
1	4	-1.441601	0.275959	0	CBSE	ISC	1
2	2	0.903724	-0.723167	0	State Board	State Board	1

Columns 9–15 of 22 · first column repeated

	GPA_1	CGPA	Current_Backlogs	...	GPA_3	GPA_4	GPA_5
0	0.964835	0.466507	3.0	...	7.56	8.47	9.79
1	0.736264	0.940191	1.0	...	6.61	9.69	6.81
2	0.202198	0.294258	1.0	...	6.53	9.40	9.81

Columns 16–22 of 22 · first column repeated

	GPA_6	Olympiads_Qualified	Technical_Projects	Tech_Quiz	Engg_Coaching	Scholarships	Misc_Tech_Events
0	9.54	0	4	2	1	0	5
1	5.25	3	2	0	0	3	4
2	6.63	3	0	0	1	3	3

3 rows × 22 columns

In [ ]:
```
### Apply One-Hot Encoding using pd.get_dummies()
```

In [14]:
```
df_241UDS0103 = pd.get_dummies(
    df_241UDS0103,
    columns=["Board_10th", "Board_12th"],
    drop_first=True
)
df_241UDS0103.head(3)
```

Out[14]:

Columns 1–8 of 22

	Branch	Marks_10th	Marks_12th	Gender	Category	GPA_1	CGPA
0	6	2.021525	-1.217335	0	0	0.964835	0.466507
1	4	-1.441601	0.275959	0	1	0.736264	0.940191
2	2	0.903724	-0.723167	0	1	0.202198	0.294258

Columns 9–15 of 22 · first column repeated

	Current_Backlogs	Ever_Backlogs	GPA_2	...	Olympiads_Qualified	Technical_Projects	Tech_Quiz
0	3.0	0.0	9.57	...	0	4	2
1	1.0	1.0	6.60	...	3	2	0
2	1.0	1.0	9.69	...	3	0	0

Columns 16–22 of 22 · first column repeated

	Engg_Coaching	Scholarships	Misc_Tech_Events	Board_10th_ICSE	Board_10th_State Board	Board_12th_ISC	Board_12th_State Board
0	1	0	5	False	False	True	False
1	0	3	4	False	False	True	False
2	1	3	3	False	True	False	True

3 rows × 24 columns

In [ ]:
```
## Step 6: Convert Boolean Datatypes into Integers
```

In [15]:
```
bool_columns_241UDS0103 =
df_241UDS0103.select_dtypes(include="bool").columns
for col in bool_columns_241UDS0103:
    df_241UDS0103[col] = df_241UDS0103[col].astype(int)
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 220 entries, 0 to 219
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  220 non-null    int64
 1   Marks_10th              218 non-null    float64
 2   Marks_12th              218 non-null    float64
 3   Gender                  220 non-null    int64
 4   Category                220 non-null    int64
 5   GPA_1                   219 non-null    float64
 6   CGPA                    219 non-null    float64
 7   Current_Backlogs        219 non-null    float64
 8   Ever_Backlogs           219 non-null    float64
 9   GPA_2                   220 non-null    float64
 10  GPA_3                   220 non-null    float64
 11  GPA_4                   220 non-null    float64
 12  GPA_5                   220 non-null    float64
 13  GPA_6                   220 non-null    float64
 14  Olympiads_Qualified     220 non-null    int64
 15  Technical_Projects      220 non-null    int64
 16  Tech_Quiz               220 non-null    int64
 17  Engg_Coaching           220 non-null    int64
 18  Scholarships            220 non-null    int64
 19  Misc_Tech_Events        220 non-null    int64
 20  Board_10th_ICSE         220 non-null    int64
 21  Board_10th_State Board  220 non-null    int64
 22  Board_12th_ISC          220 non-null    int64
 23  Board_12th_State Board  220 non-null    int64
dtypes: float64(11), int64(13)
memory usage: 41.4 KB
```

In [ ]:
```
## Step 7: Save the Processed Dataset
```

In [16]:
```
df_241UDS0103.shape
```

Out[16]:
```
(220, 24)
```

In [17]:
```
df_241UDS0103.to_csv("Experiment_8_241UDS0103_Processed_Student_Feature_Data
.csv", index=False)
print("Processed dataset saved successfully.")
```

```
Processed dataset saved successfully.
```

In [ ]:

```
## RESULT
The dataset was preprocessed by removing an unnecessary feature, scaling
numerical features with StandardScaler and MinMaxScaler, encoding
categorical features with LabelEncoder and one-hot encoding, converting
Boolean columns to integers, and saving the processed dataset.
```